The ‘Premium Brand’ Strategists

The Business Problem

- **Launching a $80 Premium Tequila — Where First?**
- **The Mandate:** Predict Average Price per Bottle by ZIP code using Census Demographics
- **The Opportunity:** Identify Early Adopter neighborhoods — which demographics drive premium buying?
- **Data:** Iowa Liquor Sales + US Census ACS 9M transactions | BigQuery

The Data

- **From 9 Million Transactions to 437 ZIP Markets**
- **Challenge:** Transactions are individual sales
  Census is ZIP code level
  = Granularity Mismatch
- **Solution:** Aggregated all transactions per ZIP 1 row per ZIP per year
- **Result:** 437 clean ZIP markets ready for modeling

The EDA

- **What Did Demographics Tell Us?**

- **Income vs Price:** correlation = -0.001
- **Education vs Price:** correlation = 0.169

- **Average price: $14.31; Std deviation: $1.97**

- **Finding:**
  -Who lives in a ZIP barely predicts
  -What they spend on liquor

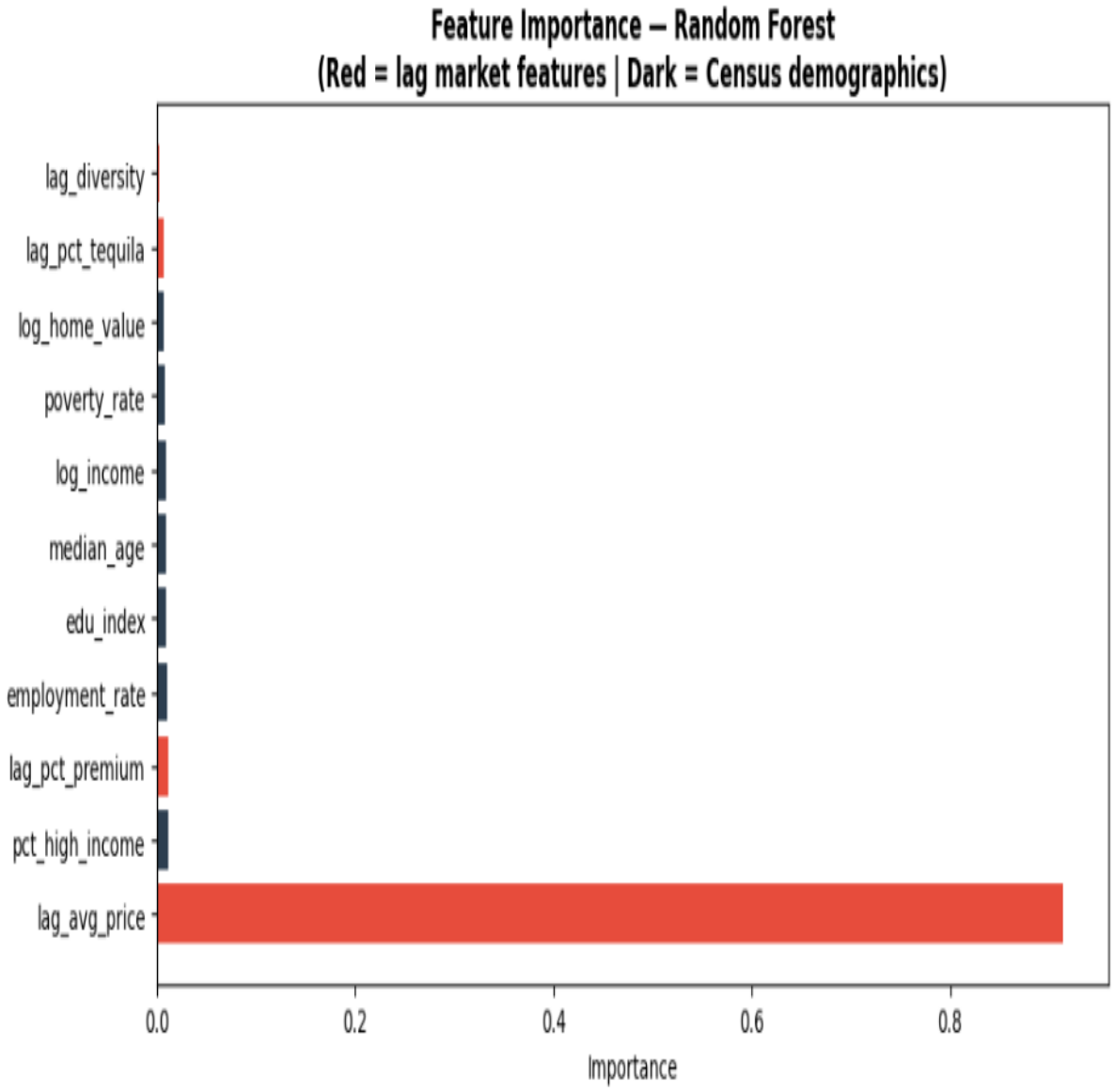
- This motivated us to look at market behavior instead

The Model

Model	RMSE	R^2	MAE
Naive Mean	2.039	-0.027	1.574
Ridge	1.018	0.744	0.750
Random Forest	1.007	0.749	0.744
Gradient Boosting	1.036	0.734	0.763

Best Model: Random Forest

Lag Features



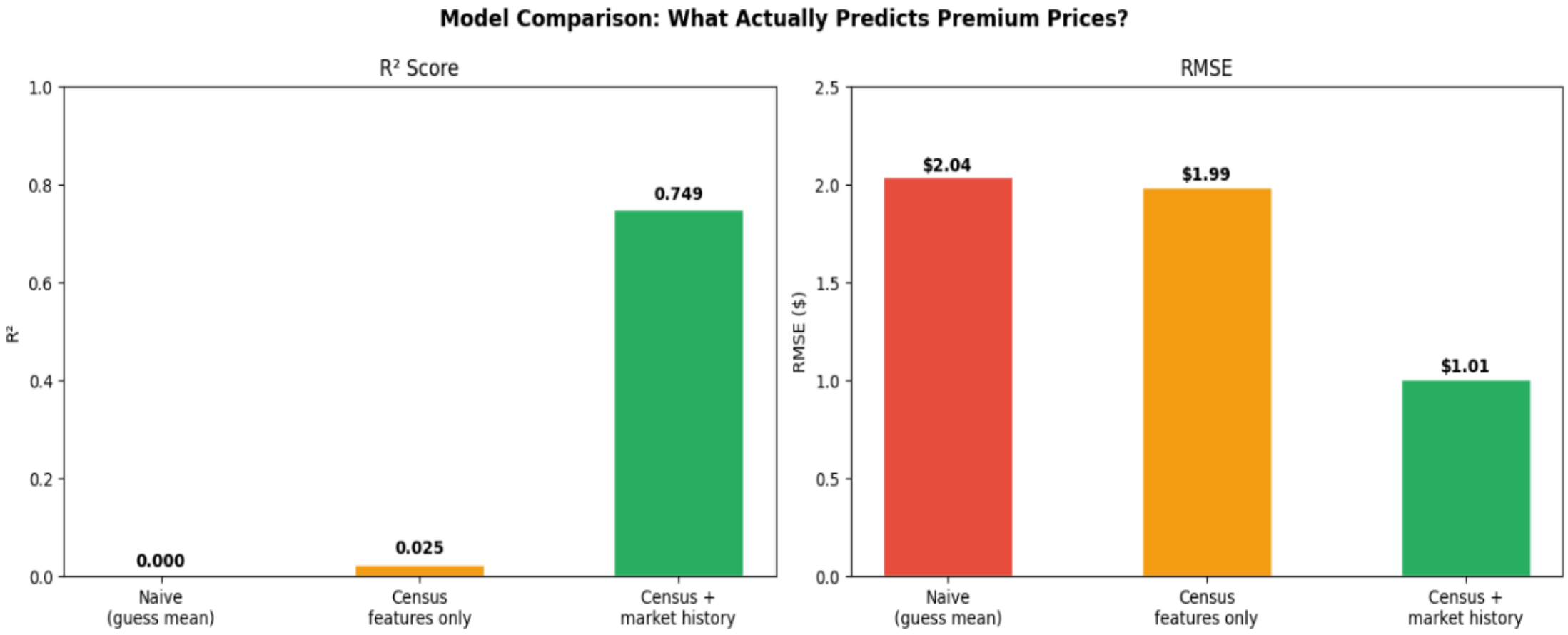
Feature	Importance
lag_avg_price	0.914120
pct_high_income	0.011655
lag_pct_premium	0.011622
employment_rate	0.010481
edu_index	0.009611
median_age	0.009426
log_income	0.008953
poverty_rate	0.008253
log_home_value	0.007187
lag_pct_tequila	0.006739
lag_diversity	0.001950

```
Census-only model (no lag features):
  R²:   0.025
  RMSE: $1.99

Full model (census + lag):
  R²:   0.749
  RMSE: $1.01

Lag features add this much value on top of census alone.
```

Model Comparison – Visual



Business Impact

- Business Impact in Dollars
- **Best model:** Random Forest
- **RMSE:** $1.01 per bottle
- **R^2**: 0.749
- **Error %:** 6.9%
- **Scenario:** 50 ZIPs x 5,000 bottles x $80

  Total revenue: $ 20,000,000

  Risk (our model): $ 1,375,521

  Risk (naive): $ 2,785,190

  Estimated Value Impact: $ 1,409,669 based on error reduction

Business Decision Layer

```
...
TOP 10 ZIP CODES — LAUNCHING HERE FIRST
```

zip_code	pct_premium	median_income	price_growth	pct_tequila	LAUNCH_SCORE
50643	0.109	85905.0	30.000	0.009	0.715
52338	0.146	101513.0	9.153	0.038	0.688
51561	0.174	58523.0	9.174	0.028	0.680
52351	0.112	105729.0	14.841	0.015	0.664
52324	0.125	93333.0	12.330	0.025	0.645
52166	0.149	46667.0	11.905	0.026	0.604
50324	0.104	64337.0	30.000	0.052	0.600
50472	0.158	54145.0	7.986	0.044	0.598
50072	0.080	71759.0	30.000	0.022	0.586
52411	0.104	127361.0	2.173	0.052	0.564

```
Why these ZIPs:
Avg premium buying:  12.6%
Avg income:          $80,927
Avg price growth:    15.8%
Avg tequila gap:     96.9%
```

How we ranked ZIPs:

- Premium adoption rate 35%
- Median income level 25%
- Price momentum 2017→2019 25%
- Tequila whitespace gap 15%

What the data showed:

- Premium buying: 12.6% vs Iowa avg 5.6%
- Income: $80,927 vs Iowa avg $54,051
- Price growth: 15.8% vs Iowa avg 3.0%
- Tequila gap: 96.9% whitespace

Verification

- **Score uses growth CAPPED at 30%**
- Verification - ACTUAL real growth numbers
- ZIP 50643 (Ranked #1)

  2019 avg price: $18.49

  2019 pct premium: 10.9%

  2019 pct tequila: 0.9%

  2019 transactions: 1,312

  2019 total bottles: 10,855

  2017 avg price: $12.12

  Actual price growth: 52.6%
- Score used: 30.0% (capped for fair ranking)

Verification for all 10 ZIPs

```
Checking for all TOP 10
Score growth capped at 30% | Actual growth
1 ZIP 50643: $12.12→$18.49 actual:52.6% scored:30.0% 1,312 transactions
1 ZIP 52338: $18.84→$20.57 actual:9.2% scored:9.2% 2,229 transactions
1 ZIP 51561: $18.42→$20.11 actual:9.2% scored:9.2% 472 transactions
1 ZIP 52351: $14.53→$16.69 actual:14.8% scored:14.8% 466 transactions
1 ZIP 52324: $16.57→$18.61 actual:12.3% scored:12.3% 432 transactions
1 ZIP 52166: $19.86→$22.22 actual:11.9% scored:11.9% 542 transactions
1 ZIP 50324: $9.42→$15.23 actual:61.7% scored:30.0% 4,936 transactions
1 ZIP 50472: $18.56→$20.04 actual:8.0% scored:8.0% 1,270 transactions
1 ZIP 50072: $12.57→$16.41 actual:30.6% scored:30.0% 274 transactions
1 ZIP 52411: $16.09→$16.44 actual:2.2% scored:2.2% 9,314 transactions
Score Check
Premium signal (0.35 weight):
  Top 10 avg: 12.6%
  Iowa avg:   5.6%
  Top 10 higher:  YES

Income signal (0.25 weight):
  Top 10 avg: $80,927
  Iowa avg:   $54,051
  Top 10 higher: YES

Price momentum (0.25 weight):
  Top 10 avg score growth: 15.8%
  Iowa avg growth:         3.0%
  Top 10 higher:  YES

Tequila gap (0.15 weight):
  Top 10 avg tequila: 3.1%
  Iowa avg tequila:   3.5%
  Top 10 lower (gap):  YES
```

The Finale

- "We did not find our customer. The data introduced us.“

- Census demographics alone: $R^2 = 0.025$
- Add market behavior: $R^2 = 0.749$

- **Decision Layer:** Top 10 ZIPs buy premium

  2.3x more than Iowa - Income 50% above Iowa avg

  Prices growing 5x faster than Iowa average